

Image generation

LESSON 07 · MODULE 2 — MULTIMODAL

What you'll learn

- How `image_model` flips a normal agent into one that returns image artifacts
- Why the `provider` (text LLM) and the `image_model` are separate roles
- How to retrieve and save generated files from the response
- How to swap the image model on an existing agent at runtime
- The aesthetic difference between DALL-E 3 (illustrative) and Gemini Pro image (literal/photographic)

Key concepts

Same agent shape, different artifact. Add

`image_model=DALLE.dalle_3` to `create_agent` and the same `agent.run()` call you've used in every video so far comes back with a file instead of text.

TAKEAWAY

Same agent shape, different artifact.

Two LLMs, two roles. The `provider` parameter is the LLM that orchestrates the request — it takes the user's words and shapes them into something the image model wants. The `image_model` does the actual rendering. You can pair a cheap orchestrator (GPT-mini) with an expensive renderer, or vice versa.

TAKEAWAY

Two LLMs, two roles.

Available image models. DALL-E 3, DALL-E 2, GPT-Image-1, GPT-Image-1.5, Gemini Pro image, Gemini Flash image — among others.

TAKEAWAY

Available image models.

Artifact retrieval is uniform. `response.has_files` tells you whether the run produced files. `response.files` is the list. Each artifact has `format_type` (image / document / audio), `name`, and `download(path)`.

TAKEAWAY

Artifact retrieval is uniform.

Swap the renderer at runtime with `agent.set_image_model()` — no need to create a new agent. Useful for A/B-ing aesthetics on the same prompt.

Aesthetic tradeoffs. DALL-E 3 leans illustrative — it embellishes, paints stylized continents on a "map," reaches for cinematic lighting. Gemini Pro image leans literal — given the same "stay literal" instruction, it produced a real cartographic map with readable place names. Neither is right; reach for DALL-E when you want art direction baked in, reach for Gemini when you want a near-photographic translation of words.

TAKEAWAY

Aesthetic tradeoffs.

Code shown in this lesson

PYTHON

```
from lyzr.image_models import DALLE

agent = studio.create_agent(
    name="image-generator",
    provider="gpt-5-mini",           # orchestrator
    image_model=DALLE.dalle_3,      # renderer
    role="creative image generator",
    goal="produce visually striking images from short prompts",
    instructions="Stay literal to the prompt.",
)

response = agent.run("A weathered brass compass over a paper map, warm dramatic lighting.")

if response.has_files:
    for artifact in response.files:
        if artifact.format_type == "image":
            path = safe_file_name(artifact.name) # helper from setup
            artifact.download(path)
            display.Image(path)
```

PYTHON

```
# Same agent, swap to Gemini at runtime
from lyzr.image_models import GeminiImage

agent.set_image_model(GeminiImage.gemini_pro) # gemini-3-pro-image-preview
response = agent.run("A weathered brass compass over a paper map, warm dramatic lighting.")
```

TRY THIS

- 1 Take a single prompt — *"A weathered brass compass over a paper map, warm dramatic lighting"* — and generate it twice on DALL-E 3 and twice on Gemini Pro image. Open the four images side-by-side and write two sentences describing the aesthetic split.
- 2 Within a single agent, use `agent.set_image_model()` to swap renderers between two consecutive calls of `agent.run()` — same prompt both times, different renderer. Save both images. Verify you didn't have to recreate the agent.